Jurnal Krisnadana (Jurnal Komputer, Sistem Kendali & Jaringan)
Vol 4, Issue 2, January 2025 , pp. 111-121
https://doi.org/10.58982/krisnadana.v4i2.739

E-ISSN: 2808-3563
P-ISSN: 2808-3520

Implementation of Grayscale Image Transformation and Histogram Equalization Methods in Digital Image Processing

Kusnadi[1*], Diyaa Aaisyah Salmaa Putri Atmaja[2]

[1*]*Teknik Informatika, Fakultas Teknik, Universitas Nahdlatul Ulama Kalimantan Timur*
[2]*Teknik Industri, Fakultas Teknik, Universitas Nahdlatul Ulama Kalimantan Timur*
[1*]*kusnadi@unukaltim.ac.id* , [2]*diyaaaisyahsalma@unukaltim.ac.id*

ARTICLE INFO

Article history:
Received 18 December 2024
Accepted 19 January 2025
Published 31 January 2025

Keywords:
Digital Image;
Grayscale Image;
Histogram Equalization

ABSTRACT

Medical digital image processing is typically color-based and exhibits varying quality, which necessitates further processing to enhance visibility and contrast. The main issue addressed in this study is how to process color images to produce higher-quality images that facilitate analysis and highlight hidden features. This research proposes several image processing methods to tackle these problems, including color image (RGB) to grayscale conversion, image inversion, constant addition, and histogram equalization. Each step aims to improve the contrast and quality of medical images, producing clearer and more informative results. Histograms are used to analyze pixel intensity distribution, while histogram equalization serves to normalize the intensity distribution and enhance image contrast. The results of this study demonstrate that by applying these techniques, medical images can experience significant quality improvement, with sharper contrast and more evenly distributed pixel intensities. This process allows previously hidden areas in dark or bright regions of the image to become clearer. The processed images help in further analysis and can improve the accuracy of medical diagnosis.

© 2025 Authors. Licensed under CC BY-SA 4.0

1. Introduction

The rapid advancement of digital imaging technology has led to an increasing need for effective image processing techniques [1]. Along with the widespread use of digital cameras, smartphones, and other imaging devices, the ability to manipulate and enhance digital images has become increasingly important in various fields. This paper focuses on two fundamental image processing methods: grayscale transformation and histogram equalization.

Grayscale image transformation is a very important technique to simplify color images while retaining important structural information. By converting RGB color images to grayscale, we can reduce computational complexity and highlight important features for further analysis [2] . This process involves carefully mapping color intensities to grayscale values to retain image contrast and detail. The advantages of using grayscale images include

reduced storage requirements, faster processing times, and simplified analysis for certain applications such as edge detection and pattern recognition [3].

Meanwhile, histogram equalization is a powerful method to improve image contrast. By redistributing pixel intensities across the available range, histogram equalization can dramatically improve the visibility of detail in both underexposed and overexposed images[4] . This technique is particularly valuable in fields such as medical imaging, where subtle variations in tissue density can be critical to diagnosis [5]. Benefits of histogram equalization include increased contrast, improved visibility of image detail, and the ability to normalize images for consistent analysis across different lighting conditions [6].

These techniques offer several advantages in image processing, including improving the quality of the image, enhancing visual appearance, making details clearer and easier to interpret [7]. In enhanced feature extraction, grayscale transformation and histogram equalization can highlight important image features, and facilitate more accurate analysis in computer vision applications [8]. Furthermore, it can reduce computational complexity; working with grayscale images often reduces processing time and memory requirements compared to color images [2]. And can normalized image representation, histogram equalization can help standardize image brightness and contrast making it easier to compare images from different sources or lighting conditions [9][10].

In this work, we explore these two techniques in depth, examining their mathematical foundations, implementation methods, and practical applications. We demonstrate how grayscale image transformation and histogram equalization can be applied to improve image quality and extract valuable information from digital images. Through a series of experiments and visual examples, we illustrate the effectiveness of these methods in various scenarios and discuss the results.

By providing a comprehensive overview of grayscale image transformation and histogram equalization, this study aims to systematically process digital images, improve image quality, and extract valuable information from digital images, thus equipping readers with an understanding of various image processing tools using these methods. These techniques serve as building blocks for more advanced image analysis and computer vision applications, making them invaluable to researchers and practitioners working with digital images.

2. Method

In this study using research methods as a reference for the stages that will be carried out by researchers, to facilitate the implementation and analysis of problems to determine solutions to research. In this study, there are several important stages in digital image processing including, providing the dataset to be tested, image conversion, histogram creation, conversion to grayscale, inversion operation, adding constants, and finally histogram equalization. The flowchart of the research method can be seen in Figure 1. dataset used is from RepoMed Nusa Mandiri University, in the form of Pap Smear medical images in JPEG format.

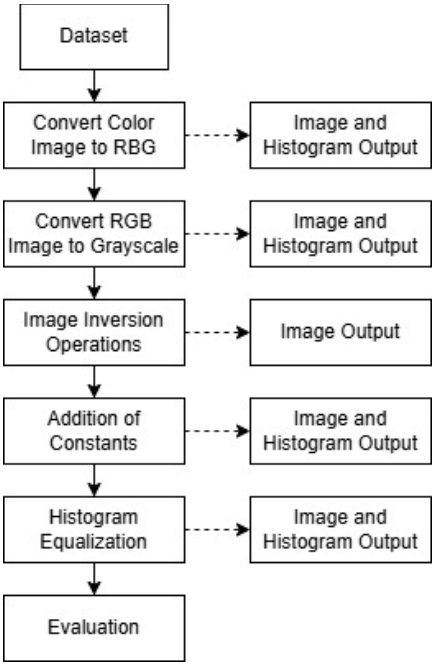

Figure 1. Flowchart of research method

There are 7 stages that will be carried out in this research, which are as follows;

1. Search and determine datasets in the form of medical images with JPEG format.
2. Perform the process of converting the color image into RBG format and histogram. The purpose of this stage is to enter the image and convert the color image to RBG along with obtaining the histogram of all pixels in the color image and the RBG channel image histogram.
3. Converting from RGB channel to grayscale image and processing the grayscale histogram. The purpose of this step is to obtain a better quality of object separation lines and the histogram output.
4. Perform an inversion process on the grayscale image. The purpose of this stage is to obtain a display on color changes, from light to dark and dark to light colors.
5. Transforming the grayscale image by adding constants and processing the histogram. The goal is to make all pixels brighter, resulting in an effect where the brightness level of the grayscale image decreases and is accompanied by the histogram output results.
6. Performing the stages of histogram equalization process on grayscale image and histogram processing, as output in the form of image and histogram. The purpose of this process is to improve image quality, increase image contrast, and evenly distribute the histogram.
7. The last stage is to evaluate the output image obtained. The purpose of this stage is to evaluate the results of the process of the grayscale image transformation method and histogram equalization in digital images.

Digital Image

A digital image is a numerical representation of a two-dimensional (2D) image. Digital images can be classified into vector or bitmap types, depending on whether the resolution is fixed or changeable [2]. Bitmap images consist of a finite number of pixels, with a fixed number of pixels in rows and columns. A pixel, the smallest individual element in an image, and stores the color description value at that moment [11].

When bitmap images are enlarged, they can appear pixelated or "broken" [12]. Typically, pixels are stored on a computer as a bitmap or raster map image in the form of a 2D array of small integers and are often transmitted or stored in compressed form [13]. A vector consists of points that have direction and length [14]. Many images combine bitmap and vector elements, such as billboards that display text (vector) and photos (bitmap) [15].

Digital images can be viewed using various software applications, including "Windows Photo Viewer". Web browsers are capable of displaying standard image formats such as GIF, JPEG, and PNG. In addition, web browsers can render SVG (*Scalable Vector Graphics*) formats [16].

Grayscale Mind

A grayscale image is an image that only displays different levels of gray color. Grayscale images consist of red, green, and blue colors with the same intensity in RGB space, so each pixel only needs one intensity value. This is different from color images that require three intensity values for each pixel [2].

Grayscale images consist of varying degrees of darkness, from deep black to bright white. Simply put, this image consists of black, white, and various shades of grey, with varying degrees of grey brightness. In grayscale images, the value of each pixel depends on the number of data bits used to represent it. Typically, grayscale values are represented by 8 bits, which is a combination of eight binary numbers that determine the value of the pixel. Therefore, the range of pixel values is from 0 to 255 (0b00000000 to 0b11111111, where "0b" indicates a number in binary format), for a total of 256 grayscale levels. If 16-bit numbers are used for pixel value representation, then the range of values becomes 0 to 65,535, with a total of 65,536 grayscale levels [17].

In digital image processing, various procedures are applied to manipulate and analyze visual data. One common technique is to convert a color image into a grayscale representation. This process is usually performed on images that use the RGB color model, which consists of three different matrix layers: the R (red), G (green), and B (blue) layers [18]. If the calculation is performed on a color image, it must be performed separately on each of these three layers. To obtain a grayscale image from the original color image, a special formula is used that combines information from all three color channels. This conversion simplifies the image data while retaining important structural and intensity information, making subsequent processing steps more efficient and less computationally intensive [19].

To convert a color image into a grayscale image, each pixel in the color image is converted into one grayscale value. This grayscale value is usually calculated using a

weighted average formula based on the intensity values of the RGB components. The commonly used formula is:

$$\text{Grayscale} = 0.3 \text{ red} + 0.59 \text{ green} + 0.11 \text{ blue} \quad (1)$$

Grayscale images are images that contain shades of gray [20]. The use of grayscale images is preferred because these images require less information compared to color images, because each pixel does not require three intensity values as in color images. In grayscale images, the gray color consists of one intensity value, namely red (R), green (G), and blue (B) [21].

The intensity in a grayscale image is stored as an 8-bit integer, providing 256 possible values ranging from 0 to 255 (0 for black, 255 for white, and the values in between represent different shades of gray) [21].

Grayscale Transformation

Grayscale transformation is a technique in digital image processing used to modify the intensity values of pixels in a grayscale image. A grayscale image is one in which each pixel represents the intensity of light from black to white, without any color. This intensity is usually represented as a value between 0 (black) and 255 (white) in 8-bit format [13] and [22].

The transformation process involves looping through each pixel in the color image and applying the above formula to calculate the grayscale value [3] and [11]. These grayscale values are then used to replace the original RGB values of the pixels. The result of this transformation is a new image where each pixel has a single grayscale intensity value. This image has only one grayscale channel compared to three (RGB) channels in a color image.

The transformation you mentioned is an example of linear contrast stretching, a technique commonly used in image processing to enhance image contrast. The linear function used in this transformation is as follows:

$$g(x) = a\, x\, f(x) + b \quad (2)$$

Information: Where f(x) is the initial pixel intensity value, a is the scale (gain), and b is the offset [2].

In grayscale transformation, there is also a contrast adjustment process that allows us to increase or decrease the contrast of the image. For example, by changing the range of intensity values, we can make the difference between dark and light areas clearer or more subtle. This transformation also allows adjusting the brightness of the image by adding a constant value to all pixels, making the image appear brighter or darker [23].

Histogram Evenness

Another technique commonly used in *grayscale* image transformation is histogram equalization. This method improves image contrast by spreading the pixel intensity values evenly across the available range [23][24].

Histogram equalization is an image processing technique used to enhance image contrast by equalizing the pixel intensity distribution [25]. The purpose of this technique is to change the intensity histogram of the input image so that the resulting histogram has a more uniform distribution. In other words, this technique makes dark areas lighter and light areas darker, thus making previously less visible details in the image more prominent. This method helps to improve the overall visibility and clarity of the image [25][26].

3. Results and Discussions

Results

a. Image Dataset

The Pap smear image dataset used in this study was obtained from the RepoMed database of Nusa Mandiri University Jakarta in JPEG format. The dataset obtained in the form of Pap Smear medical images, will then be used in this study and the image can be seen in Figure 2.

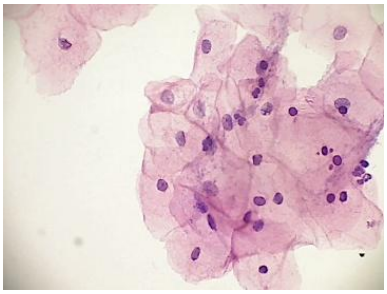

Figure 2. Pap Smear Image

b. Image Input and RBG Process

Before image importing is done, first load several tools into the library that will be used, including; numpy, pandas, cv2, skimage, PIL, and matplotlib.

In this research, it begins with image acquisition, which is the process of taking digital image files from the laptop drive that will be tested. Then the Pap Smear image conversion processing is carried out. Color Pap Smear image to image in RGB format. The image generated at the input stage is a color image so it needs to be converted to an image in RBG format. The following results of converting color images to RBG images can be seen in Figure 3a and Figure 3b.

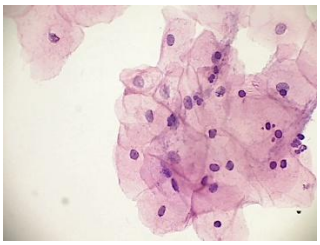

Figure 3a. Original Image

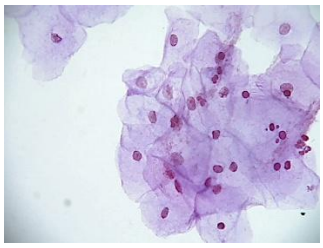

Figure 3b. RGB Image

After inputting the image and converting it to an RBG image, we perform the processing of generating the histogram of all the piskels from the color image and the histogram of the

RBG image. This process produces the histogram of all the pixels of the color image and the histogram of the RGB image. The histogram generation processing of color images and RBG images involves image contour and image histogram processing. At the beginning of the generation process, the data type of the image matrix is checked to determine the bit depth of the image. The image data obtained includes height = 2448, width = 3264, and number of *channels* = 3. Next, the processing involves image manipulation to generate a histogram of all color image pixels and a histogram of RBG channels. The results of the process produce histograms that can be seen in Figure 4, and in Figure 5, respectively.

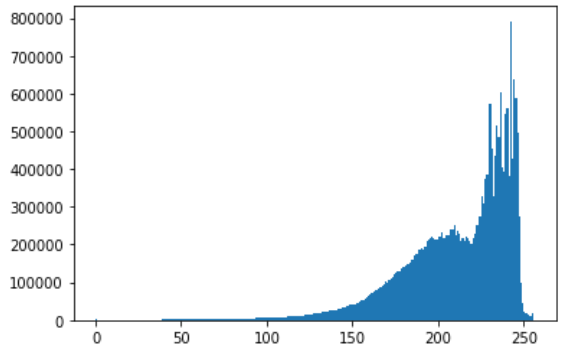

Figure 4. Histogram of all pixels in the color image

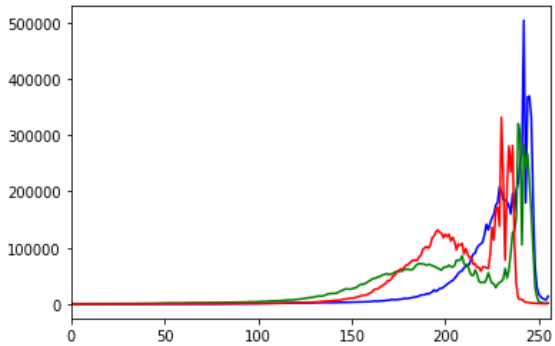

Figure 5. RGB channel histogram

c. Grayscalling Process

In this study, conversion processing and histogram generation were performed. The color medical image (RBG) becomes a grayscale image. The image produced at this stage is an RGB image so it needs to be converted to a grayscale image and at the same time produce a histogram image of grayscale image. The results of converting RGB images into grayscale images and histogram images of grayscale images can be seen in Figure 6 and Figure 7.

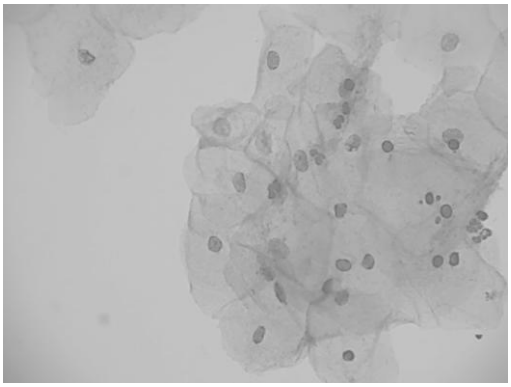

Figure 6. Grayscale Image

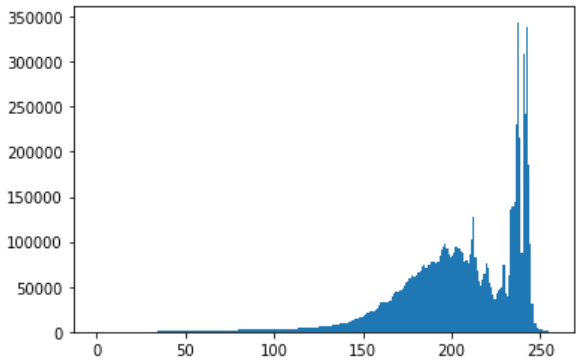

Figure 7. Histogram of the Grayscale Image

d. Image Inversion Operation Process

The image inversion operation process is performed on a grayscale color image. This image inversion operation undergoes a mathematical transformation of the grayscale image by means of a color inverse operation on the grayscale image. This stage of the process results in a change in color, from light to dark and dark to light. In grayscale images, the

intensity value of each pixel is in the range of 0 for black and 225 for white on an 8-bit scale. This operation is often used in image processing to highlight certain features or increase contrast artistically. The results of the grayscale image inversion operation process can be seen in Figure 8.

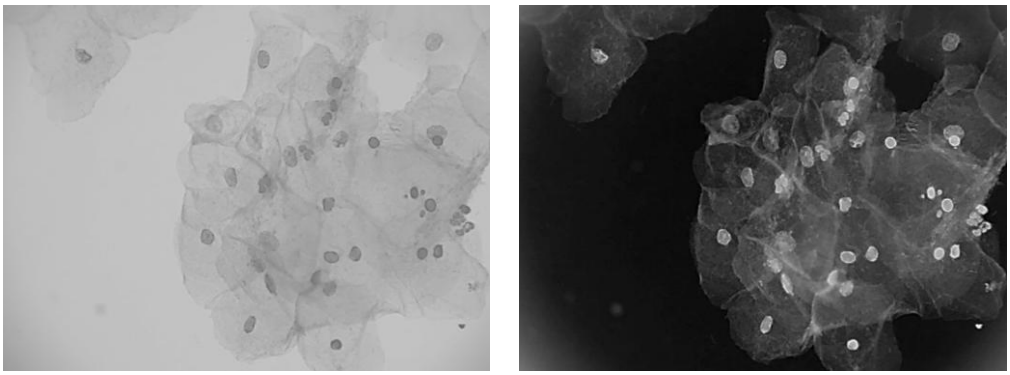

Figure 8. Grayscale Image Inversion Operation Process

e. Addition Process

In this research, the processing of adding constants to the grayscale image and its histogram. The process of adding constants to the grayscale image is done by adding twice the quality of grayscale image produced at the beginning. The result of processing with the addition of a constant twice the quality of all pixels of the grayscale image is a change in all pixels in the grayscale image to be brighter and the effect of the brightness of the grayscale image image decreases. And when the process of adding constants to the grayscale image is also accompanied by the histogram process. The images of the processing results of adding constants to the grayscale image and its histogram can be seen in Figure 9 and Figure 10.

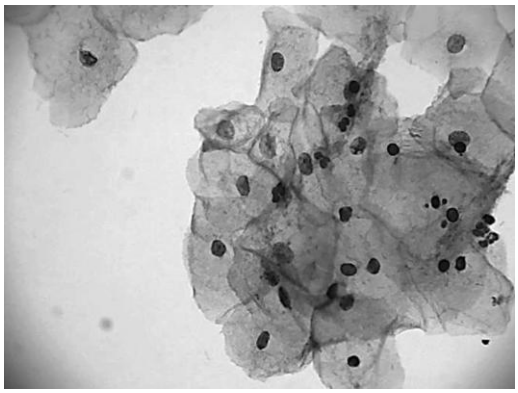

Figure 9. Grayscale image of constant addition

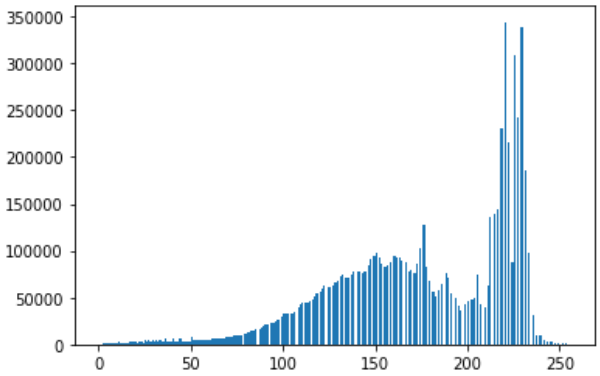

Figure 10. Histogram of constant addition grayscale image

f. Histogram Equalization Process

The last stage of this research is image processing with a histogram equalization model. This histogram equalization process serves to improve image quality, increase image contrast, and evenly distribute histograms. The image and histogram resulting from this

histogram equalization model process are clearer and sharper image quality, image contrast with even distribution of pixel intensity makes more details that were previously hidden in dark or bright areas more visible, and helps each degree of gray have a relatively equal number of pixels, and can avoid the problem of images that are too dark or too bright. The images of the grayscale image and histogram results from this histogram equalization model process are shown in Figure 11.

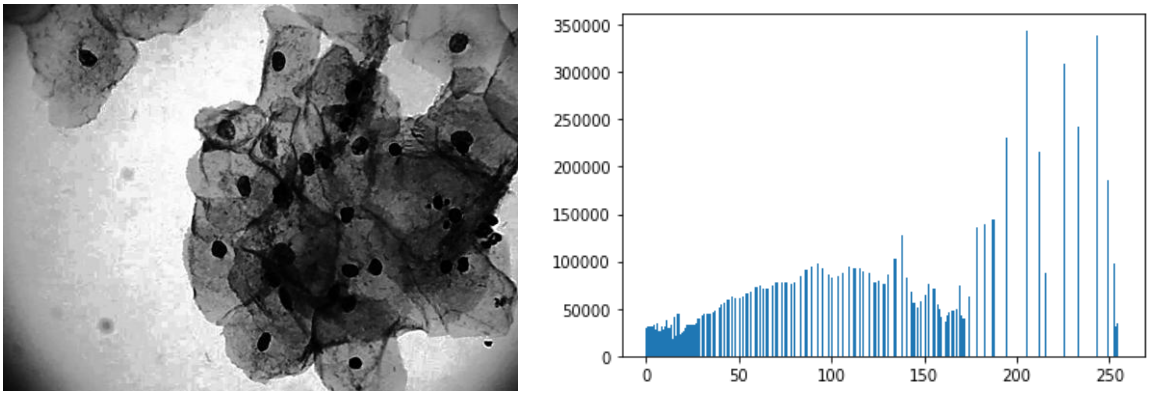

Figure 11. Grayscale Image And Histogram After Histogram Equalization

Research Discussion

This study aims to examine medical image processing techniques, especially on Pap Smear images, to improve image quality, contrast of the resulting images. Some important stages in image processing performed in this study include image conversion, histogram generation, conversion to grayscale, light inversion operation, constant addition, and histogram equalization. The following is a discussion of the results obtained at each stage of image processing:

a. **Image acquisition and conversion to RGB format:** The first process carried out in this study is image acquisition, which involves taking Pap smear image files in digital format from a laptop drive for testing. The image acquired at this stage is color and needs to be converted to RGB format for further processing. The RGB format separates the color components into three channels, namely red (Red), green (Green), and blue (Blue). The results of converting color images to RGB format are shown in Figure 3a and Figure 3b, which show the difference between the original image and the image that has been separated into RGB channels.
b. **Histogram generation from color and RGB images:** After conversion of the image to RGB, this study proceeds with the creation of a histogram of all pixels of the color image and RGB channels. This histogram serves to describe the intensity distribution of pixels in the image, be it the color image or the image separated into RGB channels. This process also involves analyzing the bit depth of the image to ensure the quality of the processed image. The image data used has dimensions of 2448 pixels high and 3264 pixels wide, with three color channels (RGB). The histogram results of the color image and RGB channels can be seen in Figure 4 and Figure 5, which show how the pixel intensity distribution varies in each color channel.

c. **RGB to grayscale image conversion and grayscale histogram generation:** The next stage is the conversion of the RGB image to a grayscale image, which converts the color image to an image in shades of gray, with intensity varying from black to white. This conversion is often done in medical image analysis to simplify information and focus attention on pixel intensity rather than color components. The result of this conversion is shown in Figure 6, while Figure 7 shows a histogram of the grayscale image depicting the intensity distribution in the grayscale image.
d. **Grayscale image inversion operation**: Next, an image inversion operation is performed on the grayscale image. This operation inverts the intensity values of the image, so that light areas become dark and dark areas become light. This technique is useful for enhancing contrast and highlighting certain features in the image. The result of the grayscale image inversion operation can be seen in Figure 8, which shows the visual difference between the grayscale image and the inversion result image, with the changes in the light and dark areas being more obvious.
e. **Constant addition to the grayscale image and its histogram:** At this stage, constant addition processing is performed on the grayscale image to increase the brightness of the image. Constant addition is done by adding twice the constant value to the previously generated grayscale image. The result of this processing is a grayscale image that becomes brighter, with an overall decreased brightness effect. This process is also accompanied by the creation of a histogram of the processed image. The results of this stage can be seen in Figure 9 and Figure 10, which show the changes in the grayscale image and its histogram distribution after the addition of the constant.
f. **Histogram equalization:** The last stage in this research is histogram equalization, which aims to improve image quality by flattening the distribution of pixel intensities across the entire range of values. This process improves the contrast of the image and spreads the pixel intensity more evenly, which helps to clarify details hidden in dark or light areas. In addition, histogram equalization also avoids the problem of the image being too dark or too light. The result of this process can be seen in Figure 12, which shows a grayscale image with sharper contrast and a more even histogram distribution, as well as improved image details that were previously not obvious.

4. Conclusion

Advances in digital image processing technology have played an important role in improving image quality by selectively conveying certain information and visual effects. Pseudo-color enhancement refers to a technology that converts grayscale images into color images through linear or nonlinear mapping functions. This research proposes a method for grayscale transformation and histogram equalization aimed at overcoming defects in image enhancement. And it shows how various image processing techniques can be applied to improve the quality and contrast of medical images, especially Pap smear images. By going through stages such as image conversion to RGB, histogram generation, conversion to grayscale, inversion operation, constant addition, and histogram equalization, the resulting image becomes clearer, sharper, and has better contrast. These techniques are very useful in medical applications to aid further analysis and improve the visibility of important features

in medical images, such as medical feature detection or diagnostic analysis. Grayscale image transformation and histogram equalization methods can be applied in future research with biomedical, biometric, and other digital image datasets to clarify images intended for analysis.

References

[1] R. Archana and P. S. E. Jeevaraj, *Deep learning models for digital image processing: a review*, vol. 57, no. 1. Springer Netherlands, 2024. doi: 10.1007/s10462-023-10631-z.

[2] R. C. Gonzalez, *Digital Image Processing*. New York: Library of Congress Cataloging in Publication data on File, 2008. [Online]. Available: www.EBooksWorld.ir

[3] M. Sonka, V. Hiavac, and R. Boyle, *Image processing, Analysis, and Machine Vision*. USA: Global Engineering, 2015. [Online]. Available: www.cengage.com/highered

[4] Y. T. Kim, "Contrast enhancement using brightness preserving bi-histogram equalization," *IEEE Trans. Consum. Electron.*, vol. 43, no. 1, pp. 1-8, 1997, doi: 10.1109/30.580378.

[5] A. M. Reza, "Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement," *J. VLSI Signal Process. Syst. Signal Image. Video Technol.*, vol. 38, no. 1, pp. 35-44, 2004, doi: 10.1023/B:VLSI.0000028532.53893.82.

[6] S. M. Pizer *et al.*, "Adaptive histogram equalization and its variations," *Computer Vision, Graphics, and Image Processing*, vol. 38, no. 1. p. 99, 1987. doi: 10.1016/s0734-189x(87)80156-1.

[7] Y. Chang, C. Jung, P. Ke, H. Song, and J. Hwang, "Automatic Contrast-Limited Adaptive Histogram Equalization with Dual Gamma Correction," *IEEE Access*, vol. 6, pp. 11782-11792, 2018, doi: 10.1109/ACCESS.2018.2797872.

[8] J. Caron and J. R. Markusen, *The Image Processing Handbook*. CRC Press Taylor & Francis Group, 2016. [Online]. Available: https://doi.org/10.1201/b18983

[9] C. Yang, M. Jin, Y. Xu, R. Zhang, Y. Chen, and H. Liu, "SepLUT: Separable Image-Adaptive Lookup Tables for Real-Time Image Enhancement," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 13678 LNCS, pp. 201-217, 2022, doi: 10.1007/978-3-031-19797-0_12.

[10] M. K. Moghimi and F. Mohanna, "Real-time underwater image enhancement: a systematic review," *J. Real-Time Image Process.*, vol. 18, no. 5, pp. 1509-1525, 2021, doi: 10.1007/s11554-020-01052-0.

[11] V. Tyagi, "Understanding Digital Image Processing," *Underst. Digit. Image Process.*, no. November, 2018, doi: 10.1201/9781315123905.

[12] C. Suresh, S. Singh, R. Saini, and A. K. Saini, "A Comparative Analysis of Image Scaling Algorithms," *Int. J. Image, Graph. Signal Process*, vol. 5, no. 5, pp. 55-62, 2013, doi: 10.5815/ijigsp.2013.05.07.

[13] S. Jamil, M. J. Piran, M. U. Rahman, and O. J. Kwon, "Learning-driven lossy image compression: A comprehensive survey," *Eng. Appl. Artif. Intell.*, vol. 123, pp. 1-14, 2023, doi: 10.1016/j.engappai.2023.106361.

[14] F. Hou *et al.*, "Poisson Vector Graphics (PVG)," *IEEE Trans. Vis. Comput. Graph.*, vol. 26, no. 2, pp. 1361-1371, 2020, doi: 10.1109/TVCG.2018.2867478.

[15] J. Cal-Gonzalez *et al.*, "Hybrid imaging: Instrumentation and data processing," *Front. Phys.*, vol. 6, no. MAY, 2018, doi: 10.3389/fphy.2018.00047.

[16] B. Sitohang and A. Sindar, "Analysis and Comparison of Sobel Edge Detection and Prewit Methods on Edge Detection of Sri Lanka Leaf Image," *J. Nas. Computing and Technol. Inf.*, vol. 3, no. 3, pp. 314-322, 2020, doi: 10.32672/jnkti.v3i3.2511.

[17] L. P. Varoslavskiy, *Digital Image Processing.*, vol. 31-32, no. 11. 1977.

[18] Savakar G. Dayanand and S. Pujar, "Digital Image Watermarking Using DWT and FWHT," *Int. J. Image, Graph. Signal Process*, vol. 10, no. 6, pp. 50-67, 2018, doi: 10.5815/ijigsp.2018.06.06.

[19] D. Riana, Kusnadi, and M. Syahrani, "Digital Image Management Using Gryascale Transformation and Histogram Equalization Methods," *J. Tech. Inform. Kaputama*, vol. 6, no. 1, pp. 108-119, 2022, [Online]. Available: https://jurnal.kaputama.ac.id/index.php/JTIK/article/view/724

[20] Y. N. Nabusa, "Digital Image Processing Comparison of Histogram Equalization and Specification Methods on Grey Images," *J-Icon*, vol. 7, no. 1, pp. 87-95, 2019.

[21] S. Yulina, "Application of Haar Cascade Classifier in Face Detection and Grayscale Image Transformation Using OpenCV," *J. Comput. Applied*, vol. 7, no. 1, pp. 100-109, 2021, [Online]. Available: https://jurnal.pcr.ac.id/index.php/jkt/

[22] A. Kaur and G. Kaur, "A review on image enhancement with deep learning approach," *Accent. Trans. Image Process. Comput. Vis.*, vol. 4, no. 11, pp. 16-20, 2018, doi: 10.19101/tipcv.2018.411002.

[23] S. Roy, K. Bhalla, and R. Patel, "Mathematical analysis of histogram equalization techniques for medical image enhancement: a tutorial from the perspective of data loss," *Multimed. Tools Appl.*, vol. 83, no. 5, pp. 14363-14392, 2024, doi: 10.1007/s11042-023-15799-8.

[24] H. Ibrahim *et al.*, "Efficient color image enhancement using piecewise linear transformation and gamma correction," *J. Opt.*, vol. 53, no. 3, pp. 2027-2037, 2024, doi: 10.1007/s12596-023-01171-4.

[25] E. Baidoo, "Implementation of Gray Level Image Transformation Techniques," *Int. J. Mod. Educ. Comput. Sci.*, vol. 10, no. 5, pp. 44-53, 2018, doi: 10.5815/ijmecs.2018.05.06.

[26] T. Trongtirakul and S. S. Agaian, "Weighted Histogram Equalization Using Entropy of Probability Density Function," vol. 1, pp. 461-464, 2021.